be less than the escape velocity from the planet if the gases should not evaporate into space. The escape velocity depends of temperature and mass of the planet (cf. §12.1.2); for example, the moon has too small a mass to keep any atmosphere. The escape velocity of H_2 and He from the Earth is too large for these elements to be retained. The original atmosphere of the Earth is therefore believed to have been mainly H_2O, CO_2, N_2, CH_4 and NH_3. This is referred to as a *reducing atmosphere*, which probably is typical for all newborn Earth-sized planets. The UV-radiation from the sun decomposes these molecules (though the fragments may recombine), releasing H_2, which escapes. The O_2 released is very reactive and combines with many elements to form oxides; most of the red surface of Mars is FeO. The common assumption is that very little oxygen was originally present as free O_2. However, rather rapidly, processes led to a shift of the atmosphere from reducing towards *oxidizing conditions*. Life appears to have begun before this transition as evidenced by fossile bacteria-like organisms in very old rocks formed in a reducing atmosphere.

12.8. EXERCISES

12.1. (a) What is the most probable kinetic energy of a hydrogen atom at the interior of the sun ($T = 1.5 \times 10^7$ K)? (b) What fraction of the particles would have energies in excess of 100 keV?

12.2. What amount (kg) of hydrogen is consumed per second by the sun in the fusion reaction 12.6 in Table 12.2? The solar constant (energy flux at earth's orbit) is 1.353 kW m^{-2} and the earth's average distance to the center of the sun is 149 599 000 km.

12.3. Assuming the sun generates its energy to 2/3 by the proton-chain and the rest by the CNO cycle, what would the expected solar neutrino flux be at the center of the earth?

12.4. In a water power station with a fall height of 20 m and a water flow of 500 $m^3\ s^{-1}$, the electric power output at 100% efficiency can be calculated. If the heavy water in the fall could be extracted and converted in a fusion reactor according to (12.14) with a 25% efficiency, which of the two power sources would yield more energy?

12.5. The supernova SN1987A was at a distance of about 51.47 kPc from us and had about 10^{46} J total energy release as estimated from the neutrino flux hitting Earth. How much mass in kg was then converted to energy in the explosion?

12.9. LITERATURE

G. Gamow and C. L. Critchfield, *The theory of atomic nucleus and nuclear energy sources*, The Clarendon Press, 1949.

E. M. Burbidge, G. R. Burbidge, W. A. Fowler and F. Hoyle, Synthesis of the elements in stars, *Rev. Mod. Phys.*, **29** (1957) 547.

H. Craig, S. L. Miller and G. J . Wassenburg (Eds.), *Isotopic and Cosmic Chemistry*, North-Holland, 1963.
Abundances of the elements in the universe, Landolt-Börnstein Physical Tables, New Series VI/1, 1965.
D. D. Clayton, *Principles of Stellar Evolution and Nucleosynthesis*, McGraw-Hill, 1968.
V. Trimble, The origin and abundances of the chemical elements, *Rev. Mod. Phys.*, **47** (1975) 877.
J. E. Ross and L. H. Aller, The chemical composition of the sun, *Science*, **191** (1976) 1223.
S. Weinberg, *The First Three Minutes*, Basic Books Inc., New York 1977.
J. Rydberg and G. Choppin, Elemental evolution and isotopic composition, *J. Chem. Ed.*, **54** (1977) 742.
The Cambridge Encyclopedia of Astronomy, Trewin Coppelstone Publ. Ltd, London 1977.
Landolt-Börnstein, New Series VI, Astronomy and Astrophysics Vol. 2a-c, Springer Verlag, 1981—1982.
P. K. Kuroda, *The Origin of the Chemical Elements*, Springer Verlag, 1982.
B. Mason and C. B. Moore, *Principles of Geochemistry*, J. Wiley & Sons, 1982.
H. D. Holland, *The chemical evolution of the atmosphere and the oceans*, Princeton Univ. Press, 1984.
H. Karttunen, P. Kröger, H. Oja, M. Poutanen and K. J. Donner, *Fundamental Astronomy*, Springer Verlag, 1987.
G. O. Abell, D. Morrison and S. C. Wolfe, *Exploration of the Universe*, Saunders College Publishing, 5th Ed., 1987.
P. A. Cox, *The Elements*, Oxford Univ. Press, 1989.
M. Roos, *Expansion of the Universe - Standard Big Bang Model*, arXiv:0802.2005v1 [astro-ph] 14 Feb 2008.

CHAPTER 13

Cosmic Radiation and Radioelements in Nature

Contents

13.1. Primary Cosmic Radiation 374
13.2. Secondary Reactions in the Earth's Atmosphere 376
13.3. Radioelements in Nature 377
13.4. Cosmogenic Radionuclides 378
13.4.1. Survey 378
13.4.2. Tritium 379
13.4.3. 14-Carbon 380
13.5. Primordial Radionuclides 381
13.5.1. Very long-lived nuclides lighter than lead 381
13.5.2. Elements in the natural radioactive decay series 382
13.6. Radium and Radon in the Environment 387
13.7. Disequilibrium 388
13.8. Age Determination from Radioactive Decay 390
13.8.1. Dating by the ^{14}C method 390
13.8.2. Dating by K-Ar method 393
13.8.3. Dating by Rb—Sr method 394
13.8.4. Dating by the Re—Os method 395
13.8.5. Dating based on ^{238}U decay 396
13.9. Natural Radioactivity of the Oceans 398
13.10. Anthropogenic Radioactivity in Nature 398
13.10.1. Nuclear weapons 399
13.10.2. Nuclear power plant accidents 399
13.10.3. Releases from nuclear power plants 402
13.10.4. Other anthropogenic additions 402
13.11. Exercises 403
13.12. Literature 404

The early workers in nuclear science found that their measuring equipment had a constant background level of radiation which could be eliminated only partially even with the aid of thick shielding walls of iron and lead. It was assumed initially that this radiation had its origin in naturally radioactive elements present in the materials in the laboratory. However, in 1911 Hess carried measuring equipment into the atmosphere with the aid of balloons and learned that some background radiation increased with altitude. Obviously, at least a component of the laboratory background radiation had its

Radiochemistry and Nuclear Chemistry
ISBN 978-0-12-405897-2, http://dx.doi.org/10.1016/B978-0-12-405897-2.00013-6
© 2013 Elsevier Inc.
All rights reserved.